

Mathematics Department

Grade 12 LS

Revision + Continuous function over an interval

I. Limits

1° When a function admits one limit , this **limit** is **unique** .

2° The properties presented in the table below help to know the limits of the functions « $f + g$ », « $f \times g$ », « $\frac{f}{g}$ » with $g \neq 0$ and « f^r » with $r \in \mathbb{Q}^{+*}$,

knowing that of f and that of g . ℓ and ℓ' are two real numbers .

The letter « a » designates either a real number or one of the symbols : $+\infty$, $-\infty$.

The question marks (?) represents the cases where we can't have an immediate conclusion . We say that these are **indeterminate forms** ; we calculate them directly , raising the indetermination, if it is possible and if their limit exists . In the other cases , the results that we admit are intuitive .

1. Limit of a sum $f + g$

If f has for limit	ℓ	ℓ	ℓ	$+\infty$	$-\infty$	$+\infty$
and if g has for limit	ℓ'	$+\infty$	$-\infty$	$+\infty$	$-\infty$	$-\infty$
then $f + g$ has for limit	$\ell + \ell'$	$+\infty$	$-\infty$	$+\infty$	$-\infty$	?

2. Limit of a product $f \cdot g$

If f has a limit	ℓ	$\ell > 0$	$\ell > 0$	$\ell < 0$	$\ell < 0$	$+\infty$	$+\infty$	$-\infty$	0
and if g has a limit	ℓ'	$+\infty$	$-\infty$	$+\infty$	$-\infty$	$+\infty$	$-\infty$	$-\infty$	$+\infty$ or $-\infty$
then $f \cdot g$ has a limit	$\ell \ell'$	$+\infty$	$-\infty$	$-\infty$	$+\infty$	$+\infty$	$-\infty$	$+\infty$	?

3. Limit of a quotient $\frac{f}{g}$

- Case where the limit of g is not zero

If f has a limit	ℓ	ℓ	$+\infty$	$+\infty$	$-\infty$	$-\infty$	$+\infty$ or $-\infty$
and if g has a limit	$\ell' \neq 0$	$+\infty$ or $-\infty$	$\ell' > 0$	$\ell' < 0$	$\ell' > 0$	$\ell' < 0$	$+\infty$ or $-\infty$
then $\frac{f}{g}$ has a limit	$\frac{\ell}{\ell'}$	0	$+\infty$	$-\infty$	$-\infty$	$+\infty$	?

- Case where the limit of g is zero

If f has a limit	$\ell > 0$ or $+\infty$	$\ell > 0$ or $+\infty$	$\ell < 0$ or $-\infty$	$\ell < 0$ or $-\infty$	0
and if g has a limit	0, remains > 0	0, remains < 0	0, remains > 0	0, remains < 0	0
then $\frac{f}{g}$ has a limit	$+\infty$	$-\infty$	$-\infty$	$+\infty$	?

4. Indeterminate forms

The cases of indeterminate forms (? in the preceding tables) need a particular study «for removing the indeterminacy». Here they are four :

$$\infty - \infty \quad ; \quad 0 \times \infty \quad ; \quad \frac{\infty}{\infty} \quad ; \quad \frac{0}{0}.$$

In each of the four above cases, the function could have a finite limit, infinite limit or no limit.

$\lim_a f$	$\lim_a g$	$\lim_a (f+g)$	$\lim_a (f \times g)$	$\lim_a \frac{f}{g}$
ℓ	ℓ'	$\ell + \ell'$	$\ell \times \ell'$	$\frac{\ell}{\ell'} \quad (\ell' \neq 0)$
$\ell \neq 0$	0	ℓ	0	$\infty \quad (1)$
0	0	0	0	?
$\ell \neq 0$	∞	$\infty \quad (1)$	$\infty \quad (1)$	0
∞	$\ell' \neq 0$	$\infty \quad (1)$	$\infty \quad (1)$	$\infty \quad (1)$
0	∞	$\infty \quad (1)$	?	0
∞	0	$\infty \quad (1)$	?	$\infty \quad (1)$
$+\infty$	$+\infty$	$+\infty$	$+\infty$	?
$-\infty$	$-\infty$	$-\infty$	$+\infty$	?
$+\infty$	$-\infty$	?	$-\infty$	?
$-\infty$	$+\infty$	?	$-\infty$	?

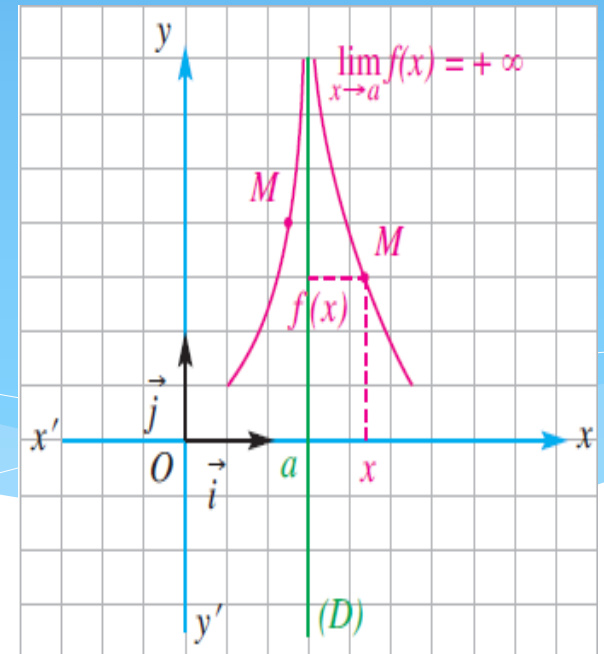
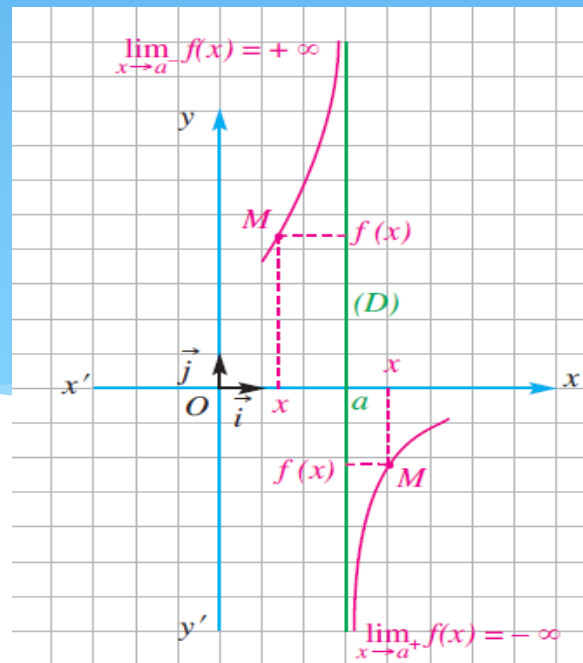
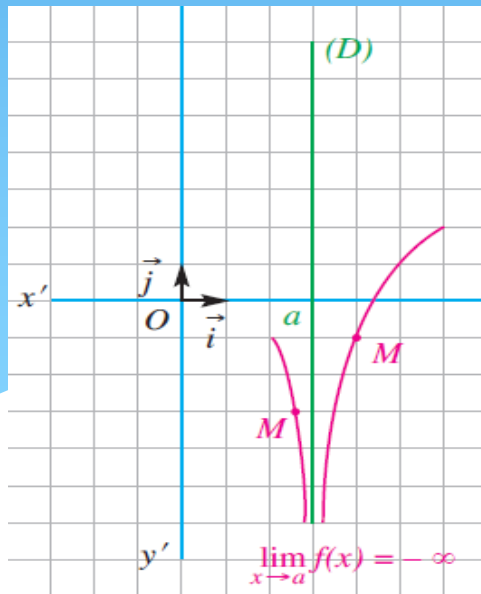
(1) : We apply the rule of signs to find if it is $+\infty$ or $-\infty$.

III. Asymptotes

1) Vertical asymptote :

Let f be a function and designate by (C) its representative curve in an orthonormal system $(O, \vec{i}, \vec{j})$.

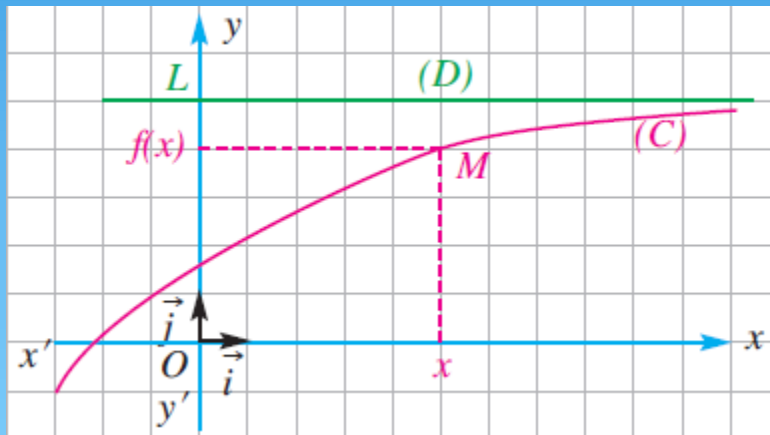
The straight line (D) of equation $x=a$ (a is a constante) is a vertical asymptote to (C) If and only if $\lim_{x \rightarrow a} f(x) = \infty$



2) Horizontal asymptote

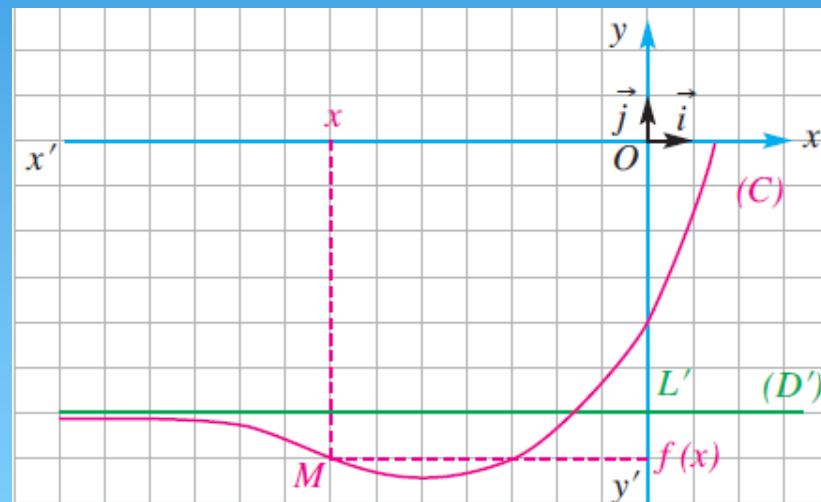
Let f be a function and designate by (C) its representative curve in an orthonormal system $(O, \vec{i}, \vec{j})$.

The straight line (D) of equation $y=b$ is a horizontal asymptote to (C) if and only if $\lim_{x \rightarrow \infty} f(x) = b$



$$\lim_{x \rightarrow +\infty} f(x) = L.$$

Then the straight line (D) of equation $y = L$ is a horizontal asymptote to (C) at $+\infty$



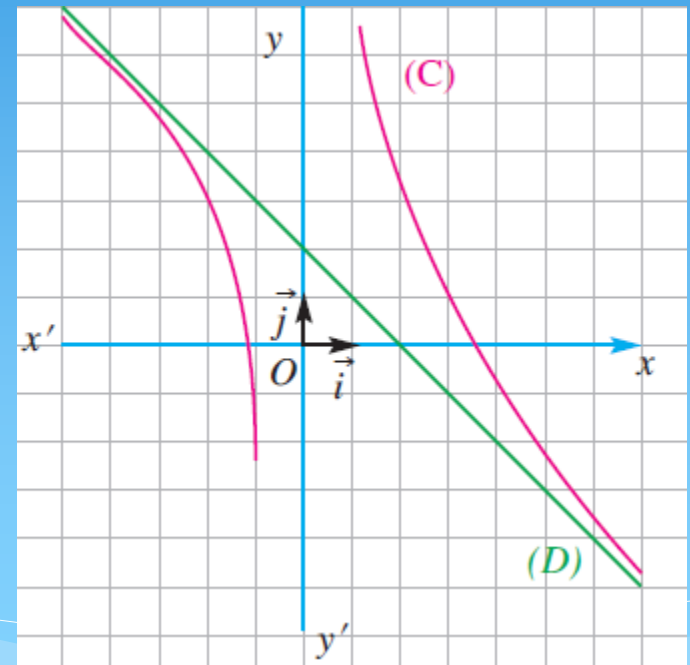
$$\lim_{x \rightarrow -\infty} f(x) = L'.$$

Then the straight line (D') of equation $y = L'$ is a horizontal asymptote to (C) at $-\infty$

3) Oblique asymptote

Let f be a function and designate by (C) its representative curve in an orthonormal system $(O, \vec{i}, \vec{j})$.

The straight line (D) of equation $y = ax + b$ is an oblique asymptote to (C) if and only if $\lim_{x \rightarrow \infty} f(x) - [ax + b] = 0$



II. Continuity

1° Definitions

a) Continuity at a point

Consider the plane of the system $(O, \vec{i}, \vec{j})$.

A numerical function f , defined over an interval containing the real number a , is said to be **continuous at a** , if and only if :

$$\lim_{x \rightarrow a} f(x) = f(a)$$

If $\lim_{x \rightarrow a^+} f(x) = f(a)$, then we say that f is continuous to the right of a .

If $\lim_{x \rightarrow a^-} f(x) = f(a)$, then we say that f is continuous to the left of a .

The continuity of f at a requires the continuity of f to the right of a and the continuity of f to the left of a : $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = f(a)$

Example

Let us study the continuity at 0 of the function f defined by :

$$f(x) = \begin{cases} x^2 + x - 1 & \text{if } x \geq 0 \\ \frac{x^2 - 1}{x + 1} & \text{if } x < 0 \end{cases}$$

f is defined at 0 and $f(0) = 0^2 + 0 - 1 = -1$.

$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x^2 + x - 1) = -1 = f(0)$; f is then continuous to the right of 0 .

$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{x^2 - 1}{x + 1} = -1 = f(0)$; f is then continuous to the left of 0 .

Then f is continuous at 0 .

Remark

If f and g are two continuous functions at x_0 , then it is the same for : $f + g$, $f \times g$, $\frac{f}{g}$ and f^r , every time these functions are defined at x_0 .

IV. Differentiability

Let f be a function defined over an interval I , Let $a \in I$ (a is a real number)

1) f is said to be differentiable at a if :

$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exist and finite

In this case the derived number of f at a is $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$

2) If $\lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}$ exist and finite we say that f is differentiable on the left of a .

In this case $f'_L(a) = \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a}$

If $\lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}$ exist and finite we say that f is differentiable on the right of a .

In this case $f'_R(a) = \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}$

f is said to be differentiable at a if: $\lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a}$

Exercise :

Let f be the fonction defined by $f(x) = \begin{cases} \frac{-2x+2}{x^2-3} & \text{si } x < -2 \\ x^2 - x & \text{si } x \geq -2 \end{cases}$

study the differentiability of f at -2

$$f(-2) = (-2)^2 + 2 = 6$$

$$= \lim_{x \rightarrow (-2)^-} \frac{\frac{-2x+2}{x^2-3} - 6}{x+2} =$$

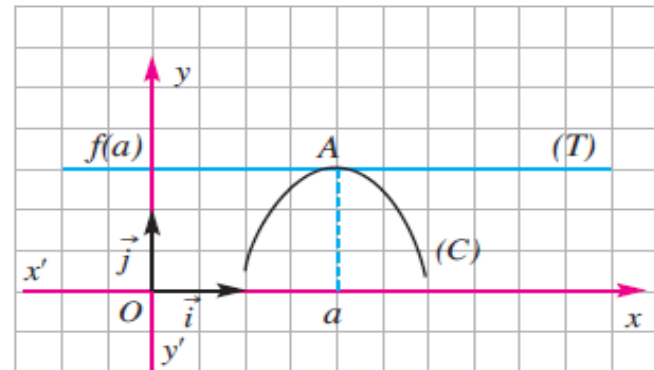
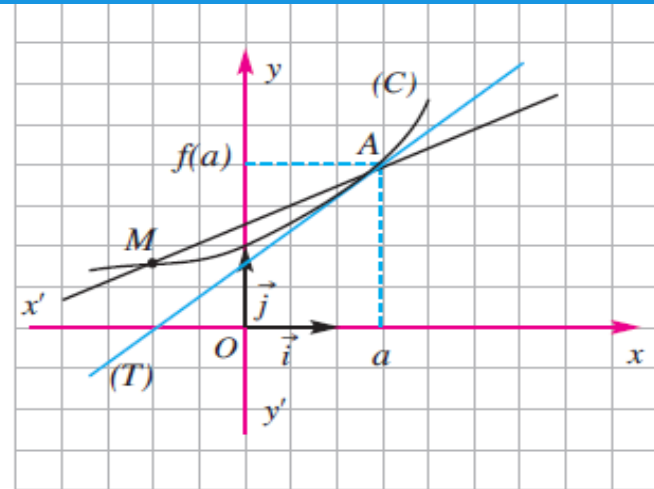
IV. Geometric interpretation of the derived number

If f is differentiable at a , then (C) admits at $A(a, f(a))$ a tangent (T) of director coefficient $f'(a)$ and of equation :

$$y - f(a) = f'(a) (x - x_A)$$

In this particular case where

$f'(a) = 0$, (T) is parallel to the abscissa axis then its equation is : $y = y_A$.



V. Table of the principal formulas of derivative

Function	Derivative function
$x \mapsto k$	$x \mapsto 0$
$x \mapsto x^n$	$x \mapsto n.x^{n-1}$
$x \mapsto a.x^n$	$x \mapsto n.a.x^{n-1}$
$a.u$	$a.u'$
$u + v$	$u' + v'$
$u.v$	$u'.v + u.v'$
$\frac{u}{v}$ with $v \neq 0$	$\frac{u'.v - uv'}{v^2}$
$\frac{1}{u}$ with $u \neq 0$	$\frac{-u'}{u^2}$
$(u)^n$	$n.(u)^{n-1}.u'$
$(au)^n$	$n.a(u)^{n-1}.u'$
$x \mapsto (\sqrt{x})$ with $x > 0$	$x \mapsto \frac{1}{2\sqrt{x}}$
$(\sqrt{u})$ with $u > 0$	$\frac{u'}{2\sqrt{u}}$

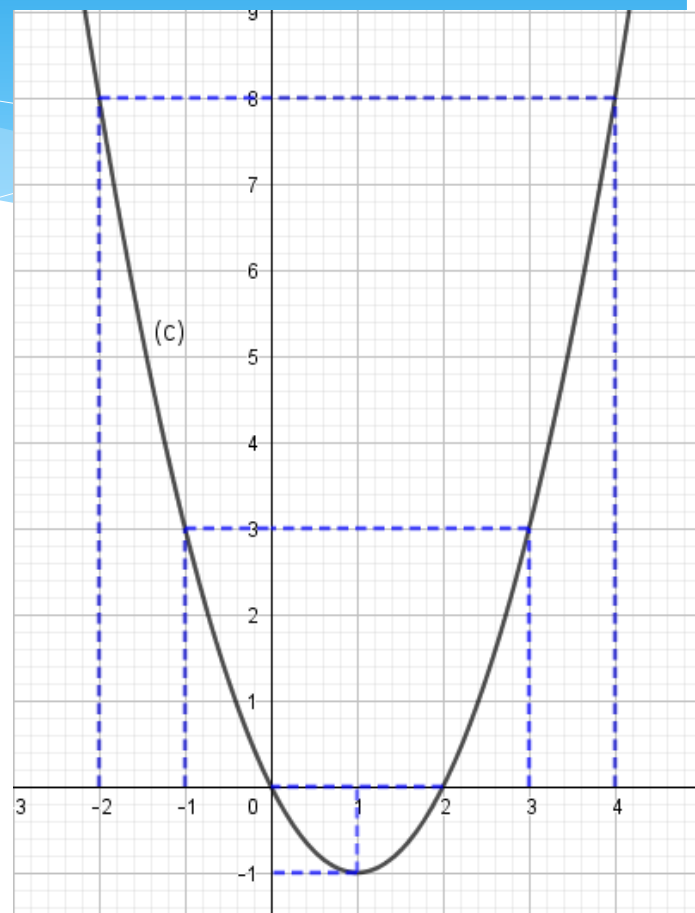
Activity

- the following curve (c) is the curve of a function f defined over $\mathbb{R}$

By graph reading

- 1) Determine $f(-2), f(-1), f(0), f(1), f(2)$
- 2) Determine the variation of f over the interval $I = [-2, 1]$, then determine the interval $f(I)$.
- 3) Determine the variation of f over the interval $I = [1, +\infty]$, then determine the interval $f(I)$.
- 4) Determine the variation of f over the interval $I = [-1, 3]$, then determine the interval $f(I)$.

Remark : $f(I)$ is an interval which contains all the values of $f(x_0)$ knowing that $x_0 \in I$



Continuous and monotonic function over an interval

Let f be a function defined over an interval I .

If f is continuous and strictly monotonic over the interval I , then $f(I)$ is an interval of the same nature as I .

Example 1:

The following curve (C) is the curve of a function f .

Using (C) f is continuous and strictly increasing over

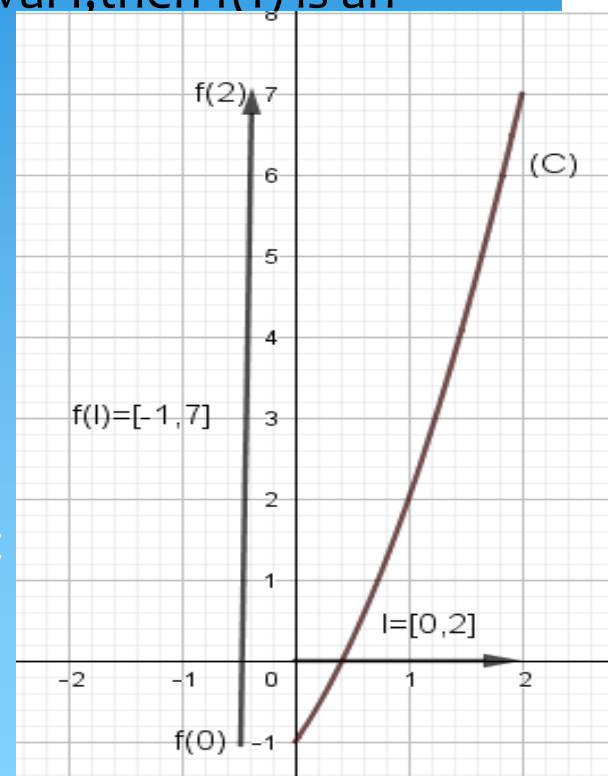
The interval $I=[0,2]$ and $f(I)=[f(0),f(2)]=[-1,7]$.

I is a closed interval and $f(I)$ is also a closed interval.

For all $x_0 \in I$ we have $f(x_0) \in f(I)$.

For all $y_0 \in f(I)$ there exists a unique $x_0 \in I$ such that

$f(x_0)=y_0$



Example 2:

The following curve (C) is the curve of a function f .

Using (C) f is continuous and strictly decreasing over

The interval $I=[-4,-1]$ and $f(I)=[f(-1),f(-4)]=[-2,7]$.

I is a closed interval and $f(I)$ is also a closed interval.

For all $x_0 \in I$ we have $f(x_0) \in f(I)$.

For all $y_0 \in f(I)$ there exists a unique $x_0 \in I$
such that $f(x_0)=y_0$

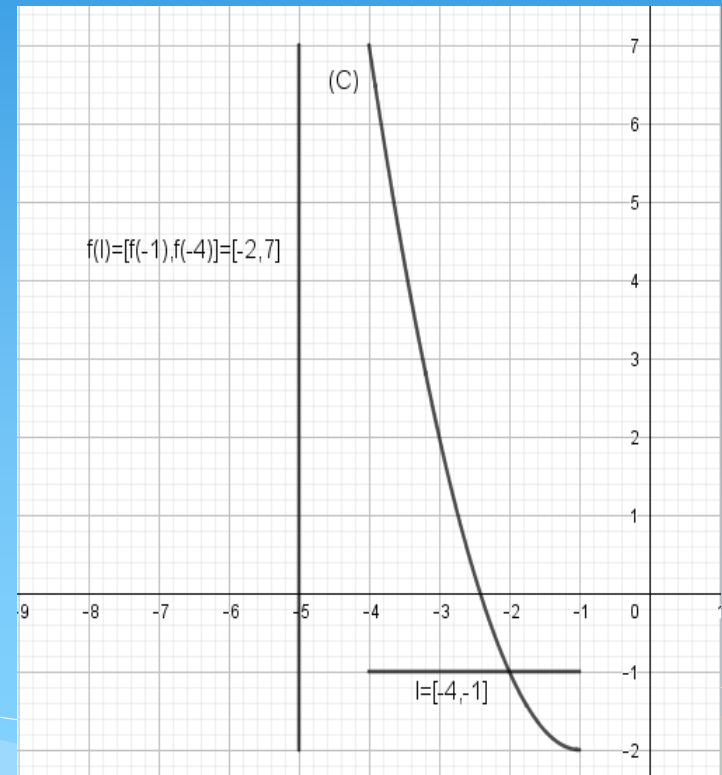


Table of possible cases

Interval I	Interval f(I) where f is continuous & increasing	Interval f(I) where f is continuous & decreasing
$I=[a, b]$	$f(I)=[f(a), f(b)]$	$[f(b), f(a)]$
$]a, b]$	$] \lim_{x \rightarrow a^+} f(x), f(b)]$	$[f(b), \lim_{x \rightarrow a^+} f(x)[$
$[a, b[$	$[f(a), \lim_{x \rightarrow b^-} f(x)[$	$] \lim_{x \rightarrow b^-} f(x), f(a)]$
$]a, b[$	$] \lim_{x \rightarrow a^+} f(x), \lim_{x \rightarrow b^-} f(x)[$	$] \lim_{x \rightarrow b^-} f(x), \lim_{x \rightarrow a^+} f(x)[$
$] -\infty, b]$	$] \lim_{x \rightarrow -\infty} f(x), f(b)]$	$[f(b), \lim_{x \rightarrow -\infty} f(x)[$
$[a, +\infty[$	$[f(a), \lim_{x \rightarrow +\infty} f(x)[$	$] \lim_{x \rightarrow +\infty} f(x), f(a)]$

Continuous function on a closed interval

If f is a continuous function on a closed interval $I=[a,b]$, then

- 1) $f(I)$ is a closed and bounded interval : $f(I)=[m,M]$
- 2) f is bounded : for every $x \in [a,b]$, we have $f(x) \in [m,M]$
- 3) f attains its bounds: there exists two values of x , $\alpha \in [a,b]$ and $\beta \in [a,b]$ such that $f(\alpha) = m$ and $f(\beta) = M$
- 4) f attains each value between m and M which means that for every $y_0 \in [m,M]$ there exists at least one $x_0 \in [a,b]$ such That $f(x_0) = y_0$

Example :

The following curve (C) is the representative curve
Of a function f defined by $f(x) = x^3 - 3x + 2$
Using the curve f is defined and continuous
over the closed interval $I = [-1.5; 1.5]$

1) $f(I) = f([-1.5; 1.5]) = [m, M] = [0, 4]$

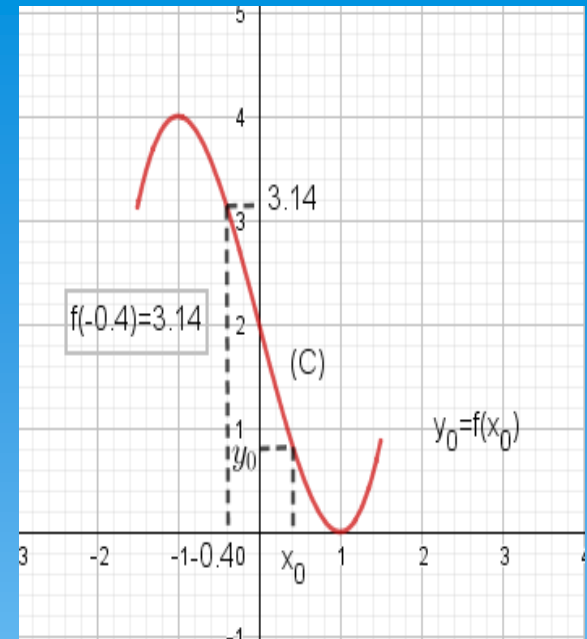
2) f attains its bounds 0 and 4 because

for $y=0 \in [0, 4]$ we have $x=1 \in [-1.5; 1.5]$ and $f(1)=0$

for $y=4 \in [0, 4]$ we have $x=-1 \in [-1.5; 1.5]$ and $f(-1)=4$

3) f attains any value belongs to $[0, 4]$

for example for $3.14 \in [0, 4]$ we have $3.14=f(-0.4)$ and $-0.4 \in [-1.5; 1.5]$



Intermediate value theorem

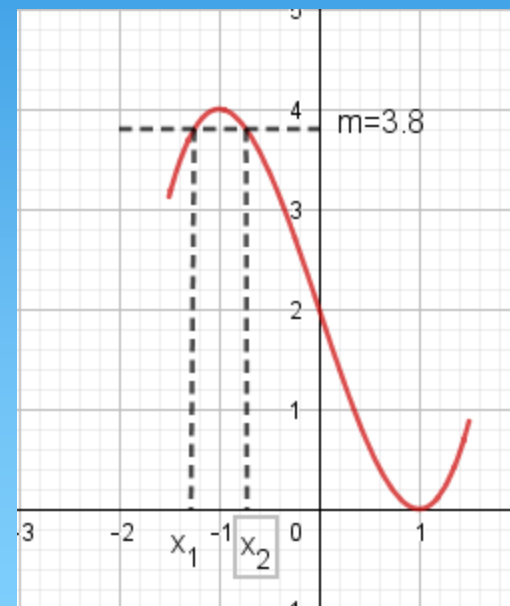
1) If f is a continuous function over a closed interval $I=[a,b]$ and if m is a real number belongs to $f(I)$ then the equation $f(x)=m$ admits at least one root $\alpha \in [a,b]$ ($f(\alpha) = m$)

In the following figure the red curve is the curve of a continuous function over $I= [-1.5; 1.5]$ and $f(I)=[0,4]$

$3.8 \in f(I)=[0,4]$ and $f(x)=3.8$ admits two roots

x_1 and x_2 that belong to $I= [-1.5; 1.5]$

$f(x_1)=3.8$ and $f(x_2)=3.8$



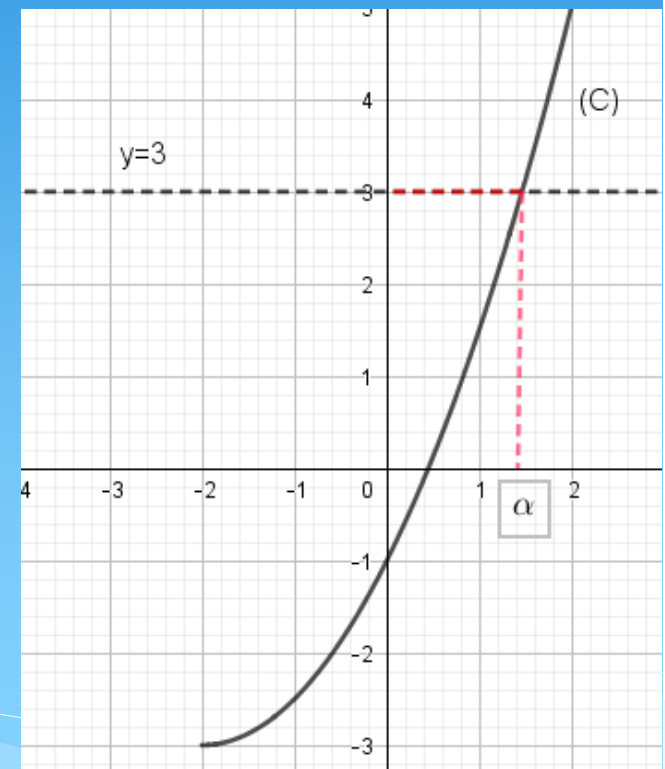
2) If f is a continuous function and strictly monotonic over a closed interval $I=[a,b]$ and if m is a real number belongs to $f(I)$ then the equation $f(x)=m$ admits a unique root $\alpha \in [a,b]$ ($f(\alpha) = m$)

To prove that $c < \alpha < d$ where $]c,d[\subset [a,b]$ we prove that:

$$[f(c)-m] \times [f(d) - m] < 0$$

if $m=0$ we prove that that $f(c) \times f(d) < 0$

Example :



Exercise

Let g be the function defined over $\mathbb{R}$ by $g(x) = 2x^3 + 5x - 4$, and designate by (Γ) its representative curve in an orthonormal system $(O, \vec{i}, \vec{j})$.

Prove that the equation $g(x) = 0$ admit a unique solution α over $\mathbb{R}$. Verify that $0 < \alpha < 1$

Solution :

g is a continuous function over $\mathbb{R}$ because it is a polynomial function .

$g'(x) = 6x^2 + 5 > 0$ then g is an increasing function over $I =]-\infty; +\infty[$

$$g(I) = g(]-\infty; +\infty[) = \left] \lim_{x \rightarrow -\infty} f(x), \lim_{x \rightarrow +\infty} f(x) \right[= \left] \lim_{x \rightarrow -\infty} 2x^3, \lim_{x \rightarrow +\infty} 2x^3 \right[=]-\infty; +\infty[$$

We have

1) g is a continuous and monotonic (increasing) function over I .

2) $0 \in g(I)$

Using I.V.T the equation $g(x) = 0$ admits a unique root $\alpha \in I$

Verify that $0 < \alpha < 1$

$$g(0) \times g(1) = -4 \times 3 = -12 < 0 \text{ so } 0 < \alpha < 1$$